

DUAL-METRIC TRADE-OFF COMPARISON OF SELECTED LITERATURE

Overview of the performance metric. ACC: Accuracy, SEN: Sensitivity, SPE: Specificity, PRE: Precision, AUC: Area under Curve, IoU: Intersection of Union, DC: Dice Coefficient, Param: Parameters, FLOP: Floating-point Operations Per Second

Ref.	Model / Architecture Focus	ACC	SEN	SPE	F1-Score	AUC	IoU	DC	Param (M)	Size (MB)	FLOP	QA Score
[1]	SkinLesionNet	94.3	92.8	-	93.0	-	-	-	-	14.0	-	3.0
[2]	kCC-Net	91.0	78.6	96.7	-	-	-	-	0.61	-	-	2.5
[3]	ECA-MobileNetV2 model	82.3	-	-	-	96.9	-	-	3.34	-	319.9M	3.0
[4]	MobileNetV3-Small (KDE-CT)	93.5	-	-	-	-	-	-	1.50	-	-	2.5
[5]	CNN	98.0	-	-	-	-	-	-	10.01	-	-	2.5
[6]	Fine-tuned EfficientNetB7	92.0	88.0	-	91.0	-	-	-	66.7	256	-	4.0
[7]	MobileNetV2 with CA mechanism (MbV2SCA)	83.9	82.4	85.3	-	-	-	-	0.97	67.4	-	3.5
[8]	DCNN with MWNL-CLS	-	-	98.0	-	98.5	-	-	19.4	-	3.2B	3.5
[9]	T-Net	-	-	-	92.8	-	87.0	-	0.03	0.11	-	3.0
[10]	Melanlysis	94.0	92.5	89.5	93.0	-	-	-	-	23.2	-	3.5
[11]	MobileUNet	96.1	89.1	98.1	-	-	80.1	88.4	8.05	92.8	-	4.0
[12]	Lightweight Convolutional Neural Network (LWCNN)	99.3	-	-	99.3	99.8	-	-	1.61	-	-	3.5
[13]	MobileNet	86.1	86.0	-	86.0	-	-	-	4.25	-	-	2.5
[14]	Optimized CNN with XAI	78.8	-	-	-	-	-	-	5.19	-	-	2.5
[15]	Modified SqueezeNet	97.0	97.2	-	97.2	-	-	-	0.14	-	-	3.5
[16]	LF-Net	96.7	92.8	97.2	-	-	84.5	90.9	6.43	-	0.84G	4.0
[17]	MobileNetV2	83.9	81.7	-	82.5	-	-	-	-	9.16	-	3.5
[18]	LWCNN	98.9	-	-	98.9	99.7	-	-	1.61	-	-	3.5
[19]	Lightweight Dermoscopic Lesion Segmentation Net	-	-	-	-	-	84.0	91.0	0.35	-	1.54G	4.0
[20]	Multi-level feature fusion and attention mechanism	92.6	-	98.5	92.5	-	-	-	4.58	-	1420M	4.0
[21]	Ultra-Lightweight Fully Asymmetric Convolutional Network (ULFAC-Net)	96.5	92.0	97.6	-	-	84.7	90.9	0.84	-	3.71G	4.0
[22]	Lightweight CNN	89.7	89.1	-	89.2	93.9	-	-	4.67	-	-	3.5
[23]	CNN	97.6	-	-	-	-	-	-	1.31	-	-	3.0
[24]	Channel Attention HalfU-Net	96.0	92.6	96.5	-	-	80.5	87.9	0.034	-	-	2.5
[25]	Modified MobileNet	99.4	-	-	-	-	-	-	4.20	17.0	-	3.0
[26]	Residual attention yolov4 (RAYOLOv4)	-	-	-	86.7	-	-	-	-	128	-	3.5
[27]	CNN (Teacher and Student Models)	98.4	-	-	98.0	-	-	-	0.32	-	-	3.5
[28]	Hybrid ANN + XGBoost	94.8	94.9	94.7	94.8	95.2	-	-	6.30	250	-	3.5
[29]	VRES-CNN	98.0	-	-	-	-	-	-	0.08	0.30	31.27M	4.0
[30]	EdgeNeXt-ASCEND	92.6	-	98.5	92.5	-	-	-	5.30	-	972M	3.0
[31]	Improved EfficientNet B0	90.4	86.0	-	87.0	-	-	-	5.19	-	-	3.0

[32]	Resident Vision Transformer	94.7	94.0	-	95.00	-	-	-	8.47	11.01	-	3.0
[33]	Improved YOLOv5s	-	-	-	-	-	-	-	2.20	4.60	-	3.5
[34]	LiwTERM (ViT + LM)	-	66.0	-	-	-	-	-	66	-	-	3.5
[35]	DenseNet-201	96.0	-	-	-	-	-	-	19.31	-	-	2.5
[36]	Lightweight multi-fusion network (LMFUNet)	95.0	89.8	96.6	-	-	81.3	89.7	0.10	-	0.106G	4.0
[37]	HierAttn-s	96.7	-	99.5	96.7	-	-	-	2.14	-	1.32B	3.5
[38]	EfficientNet B0	90.0	93.0	-	92.2	-	-	-	0.16	-	-	2.5
[39]	Hybrid CNN-Transformer student model	-	-	-	-	84.6	-	-	-	9.43	-	3.5
[40]	ReXNet-150	86.4	86.3	-	86.3	-	-	-	-	30.20	-	3.0
[41]	EfficientNetB4 + FFN	91.0	88.5	-	88.0	98.8	-	-	17.56	-	-	2.5
[42]	Proposed CNN + SE	94.1	-	-	-	96.5	-	-	2.8	-	120M	3.5
[43]	MobileNet-V4	97.6	-	-	-	-	-	-	8.44	-	-	2.5
[44]	GhostNetV2 backbone with SimAM and SEAttention mechanisms	-	83.2	-	-	-	-	-	-	-	10.6G	2.5
[45]	EfficientNetV2B0+ CBAM	97.0	-	-	-	-	-	-	7.1	-	-	3.5
[46]	Lite-CNN	96.0	-	-	-	-	-	-	38.04	145.13	-	3.5
[47]	YOLOv5 + SimAM and SEAttention	-	75.4	-	-	-	-	-	-	-	16.0G	2.5
[48]	K-Means quantization + CNN	97.0	97.1	-	97.0	-	-	-	-	1.137	-	2.5
[49]	Custom CNN Student Model	85.5	-	-	-	-	-	-	2.8	-	-	3.5
[50]	Compact CNN-Transformer (CCT)	85.0	85.0	-	85.0	-	-	-	4.0	-	-	3.5
[51]	EffiDerm	93.1	-	-	-	-	-	-	1.4	5.6	-	3.5
[52]	ResNet50 + TL	89.1	-	-	-	-	-	-	23.66	-	4.0B	3.0
[53]	Deep Feature Fusion U-Net (DFF-UNet)	96.9	93.5	98.0	-	-	88.6	94.0	0.19	-	0.109G	4.0
[54]	A-Mel	99.1	-	-	-	-	-	-	0.232	1.01	-	4.0
[55]	YESnet	96.0	90.9	-	-	-	86.1	92.4	2.9	-	-	3.0
[56]	UNet + THAPRA pooling method	-	-	-	-	-	80.3	88.9	0.27	-	0.80G	4.0
[57]	UCMViT	97.2	91.9	98.5	-	-	-	92.7	-	-	2.514G	2.5
[58]	Ensemble Teacher Model + Dense + Channel-Wise Attention + CNN (Student)	98.6	-	-	99.0	-	-	-	0.32 (Student Model)	-	-	3.0
[59]	AlexNet (Pruning)	97.5	95.8	97.9	93.9	-	-	-	5.5	-	-	2.5
[60]	MPiFormer	-	-	-	72.6	-	84.7	-	5.2	-	1.667G	3.0
[61]	Sub-Sampling Aware Channel Attention (SACA)	-	98.0	-	-	-	93.0	96.0	0.34	-	1.06G	2.5
[62]	CAViT-Net	99.2	-	-	-	-	-	-	34.5	-	5.8G	2.5
[63]	ViT-B/32 combined with BERT-encoded skin tone using Element-Wise Fusion	82.3	80.2	84.6	82.20	82.4	-	-	198.25	-	-	3.0
[64]	TokenCutSeg	-	-	-	-	-	80.9	-	45.82	-	129.81G	4.0
[65]	Swin-RSCBNet	-	-	-	-	-	84.8	91.8	33.80	-	-	3.5

[66]	DSNET	-	-	-	-	-	81.3	89.8	0.291	-	0.287G	4.0
[67]	EDA-ResNet50	93.2	94.0	92.5	93.2	-	-	-	25.5	97.34	7.95	3.0
[68]	CNN-RNN	92.0	-	-	-	-	-	-	9.2	-	-	2.5
[69]	Cross-Mix Attention Module (CMAM)	96.5	-	-	-	-	84.1	90.7	4.6	-	16.85G	3.0
[70]	FBM-UNet	-	-	-	-	-	92.4	96.0	1.8	-	0.76G	4.0
[71]	KD-NAS	82.0	-	-	-	-	-	-	4.97	19.00	-	2.5
[72]	HierAttn	96.7	-	-	-	-	-	-	2.14	-	1.32B	3.0
[73]	Improved U-Net	-	93.8	97.8	-	-	-	-	12.4	-	3.8G	2.5
[74]	AMF-UNet	-	-	-	-	-	78.4	86.4	0.43	-	2.78G	3.5
[75]	MobileUNETR+	96.2	86.7	-	-	-	84.2	87.6	5.52	-	4.26G	3.0
[76]	FUSCANet	92.8	87.4	98.0	87.8	-	-	-	5.27	-	1.47G	3.5
[77]	MedLiteNet	-	-	-	-	-	-	90.5	3.2	-	9.2G	4.0
[78]	DermaScanAI	93.2	91.1	-	91.7	96.2	-	-	9.6	-	1.4G	4.0
[79]	QEMPT	89.4	-	-	87.0	-	-	-	12.6	-	-	2.5
[80]	LCS-Net	-	-	-	-	-	80.9	89.4	0.07	-	-	3.5
[81]	LiteMobileV3	99.4	99.4	-	99.4	-	-	-	4.29	16.38	0.21G	3.0
[82]	FIF-Net	-	-	-	-	-	80.1	89.0	0.05	1.2013	0.062G	4.0
[83]	VFFM-UNet	94.9	92.9	95.9	-	-	83.5	91.0	0.35	-	0.494G	3.5
[84]	Xception-based CNN with Federated Learning	98.9	99.0	-	99.0	-	-	-	-	42.8	-	3.5
[85]	CBAMSNet	-	-	-	-	-	-	-	0.112	-	0.0566G	4.0
[86]	AMST-Net	92.1	-	-	-	-	81.5	88.5	32.0	-	-	2.5
[87]	MedMamba-ESS	96.9	96.8	98.1	96.6	99.6	-	-	14.3	-	2G	2.5
[88]	CA-SAUNet	-	-	-	-	-	84.7	91.1	33.0	68.0	-	3.0
[89]	TRM-ViT	72.8	-	-	-	79.5	-	-	2.15	-	-	3.5
[90]	MAAN	94.2	91.5	-	-	-	85.7	92.1	0.37	-	0.23G	4.0
[91]	GISLC	98.2	-	-	98.2	-	-	-	12.9	-	-	3.0
[92]	HCHS-Net	97.8	97.8	99.6	97.7	99.0	-	-	0.25	-	-	3.0
[93]	SCGViT	-	-	-	97.3	-	-	-	0.524	-	0.866G	4.0
[94]	LiwTERM-r	-	66.0	-	-	-	-	-	66.0	300	-	4.0
[95]	ResNet-50	91.4	88.7	-	90.0	97.2	-	-	25.6	102	-	3.0
[96]	ELM with HOG and PCA	93.5	-	-	92.0	96.0	-	-	-	860	-	3.0
[97]	IR-MHAtt	98.6	-	-	-	-	-	-	1.02	-	0.94G	4.0
[98]	SvANet	-	93.5	-	-	-	87.5	93.2	177.6	-	-	2.5
[99]	HALF-MAFUNET	94.7	83.1	98.0	87.7	-	80.2	-	0.54	-	7.944G	3.5
[100]	MedNeXt	98.1	97.7	-	98.3	99.7	-	-	10.72	-	1.90G	3.0
[101]	EfficientSkinTrans	97.2	91.2	-	89.3	98.3	-	-	31.11	-	2.18G	2.5
[102]	CFCSE-Net	98.1	-	-	-	-	92.3	95.9	0.91	-	5.07G	2.5
[103]	GhostNetV2	95.0	94.0	-	94.0	99.0	-	-	6.1	-	167M	3.5
[104]	LMSAUnet	92.6	-	-	-	-	87.6	87.9	0.39	-	1.06G	4.0
[105]	U-Shape Architecture + DCPBAM + MFCAM	80.7	77.4	-	78.9	93.6	-	-	4.38	8.3	8.61G	4.0
[106]	Edge-Avg	93.9	93.9	-	93.9	-	-	-	0.1	-	-	3.0
[107]	HCViT-Net	96.2	85.1	98.6	-	-	87.8	93.3	5.76	-	7.51G	3.5
[108]	Modified EfficientNet-B0	99.6	-	-	-	-	-	-	4.9	24.2	-	3.5

	Optimized with Quantum-Behaved Algorithm											
[109]	MSAMamba-UNet	-	-	-	-	-	85.5	92.2	0.056	-	0.069G	4.0
[110]	VSGNet	95.9	90.0	97.5	-	-	82.5	90.4	0.15	-	0.18G	3.5
[111]	APNet	99.6	100	99.2	-	99.9	-	-	0.07	-	0.42G	3.5
[112]	ConvNeXt-Base encoder and a U- Net-style decoder	-	95.2	-	93.6	-	88.0	-	92.67	-	0.092T	3.0
[113]	Dermo-Transfer	98.5	99.0	93.5	98.0	-	-	-	-	14.1	96.9M	2.5
[114]	ConvNeXtXLarge	87.6	-	-	75.8	-	-	-	366.1	-	-	3.0
[115]	VML-UNet	-	-	-	-	-	84.5	91.2	0.53	2.18	1.24G	4.0
[116]	MetaFormer architecture	95.0	92.1	-	93.3	-	-	-	35.01	-	-	3.5
[117]	MobileNetV2	88.3	-	-	-	98.0	-	-	3.5	16.0	0.31G	4.0
[118]	SM-UNet	-	-	-	-	-	82.5	90.2	0.02	-	-	3.5
[119]	MobileNetV2	91.4	83.0	93.0	82.0	-	81.0	-	3.5	-	-	3.0
[120]	JILDYA-Net	98.8	95.9	99.3	-	98.6	-	-	2.3	-	-	3.5
[121]	NASNET Mobile	99.3	99.3	-	99.2	-	-	-	4.41	-	-	2.5
[122]	JLU- Net	-	91.9	97.6	92.0	-	85.0	91.9	0.3	-	0.54G	3.0
[123]	SkinIncept	96.5	-	-	96.3	-	-	-	49.96	-	-	2.5
[124]	CACBL-Net	96.9	90.6	97.5	-	98.9	-	-	3.8	-	-	3.5
[125]	SKINC-NET	98.5	98.6	-	98.6	-	-	-	0.67	7.7	-	4.0
[126]	ConvNeXtV2 blocks + focal self- attention mechanisms	93.6	90.1	-	90.7	-	-	-	36.44	-	-	3.5
[127]	CNN	98.3	98.0	-	98.0	-	-	-	-	0.976	-	3.5
[128]	SLP-Net	93.9	89.3	95.4	-	-	80.6	88.2	0.2	-	2.30G	4.0
[129]	D2LFS2Net	94.4	94.4	-	94.4	-	-	-	5.3	-	-	2.5
[130]	AIR-CNN	91.6	91.6	91.5	91.6	-	-	-	1.1	-	-	3.5
[131]	DDIM	-	84.5	-	-	-	-	88.2	-	84.6	-	3.0
[132]	YOLOv8 based segmentation with ViT classification	77.1	77.1	-	76.0	-	-	-	22.0	-	-	3.5
[133]	ScNet	97.1	97.1	-	97.1	-	-	-	3.53	-	0.23G	4.0
[134]	MobileUNETR	96.6	93.0	96.9	-	-	87.5	92.8	3.0	-	1.3G	3.5
[135]	EfficientFormerV2 + Swin Tiny Transformer	89.0	-	-	88.0	98.1	-	-	29.0	-	-	3.0
[136]	Unet+DPN68	92.5	80.4	-	80.5	-	68.7	80.5	16.95	-	5.18G	4.0
[137]	DeepSkinSeg	94.6	95.1	93.0	96.4	-	93.0	96.4	4.84	-	-	3.5
[138]	DenseNet169	-	93.9	-	93.6	-	-	-	13.3	-	8.0G	4.0
[139]	DermalNet Transformer	89.3	-	-	88.8	93.9	-	-	6.64	26.58	-	3.0
[140]	MobileNetV2	92.3	-	-	-	-	-	-	-	9.89	-	2.5
[141]	MSNet	92.9	-	-	-	-	84.1	-	1.735	-	17.01G	4.0
[142]	LGCPE-UNet	-	-	-	-	-	79.6	88.7	0.058	-	0.112G	3.0
[143]	MobileNetV3 + EfficientNet-Lite	94.6	94.2	-	93.8	-	-	-	-	3.6	-	3.0
[144]	DEMSAG-Net	-	-	-	-	-	83.4	90.8	2.72	-	2.42G	2.5
[145]	LSSF-Net	96.4	94.3	93.2	-	-	89.1	93.8	0.81	-	3.1G	4.0
[146]	LightDPH-B3	82.0	69.0	96.9	-	-	-	-	13.3	-	2.04G	4.0
[147]	EfficientNetB7	88.0	86.9	-	86.6	94.2	-	-	7.8	-	11.3B	3.0
[148]	MixNets	94.3	93.5	-	93.4	-	-	-	0.058	29.0	-	2.5

[149]	MABSCNET	78.2	84.7	-	78.7	-	-	-	20.14	-	-	2.5
[150]	UCM-Net	-	-	-	-	-	81.3	88.5	0.05	1.191	0.047G	4.0
[151]	LViT	95.8	-	-	-	-	-	-	60	-	-	3.0
[152]	MFLUnet	93.7	90.2	95.1	-	-	80.1	88.9	0.038	-	0.05G	3.0
[153]	Learning-efficient PCA	-	-	-	-	-	88.2	93.7	0.026	-	0.055G	3.0
[154]	LW-MHFI-Net	96.1	96.5	-	91.8	-	95.0	-	9.3	-	-	2.5
[155]	ELA-Net	92.2	-	-	-	-	82.9	-	0.459	-	8.43G	4.0
[156]	CNN	84.0	40.0	-	51.0	64.0	-	-	-	66.0	-	2.5
[157]	DTONet	93.9	-	-	-	-	83.5	89.1	0.031	-	0.126G	3.5
[158]	LA-CapsNet	98.0	98.8	68.0	98.9	99.0	-	-	1.2	-	0.36G	4.0
[159]	SqueezeNet	85.2	41.6	95.3	51.6	-	-	-	1.24	5.2	-	3.0
[160]	CR-Conformer	83.6	-	-	-	-	-	-	11.17	-	-	2.5
[161]	LUT-SLS	95.6	88.0	-	-	-	80.6	88.9	2.06	-	-	2.5
[162]	TinyStudent (ResNet50)	85.5	-	-	85.5	-	-	-	0.35	-	-	3.5
[163]	DGLA-ResNet50	90.7	-	-	-	-	-	-	104.2	-	15.6G	4.0
[164]	DSS-MLP + MSSEA	-	-	-	-	-	85.0	-	0.33	-	15.98G	3.5
[165]	Resnet50/MobileNet V2 with Wavelet- based Feature Compression	85.6	-	-	80.5	95.6	-	-	3.5	-	-	2.5
[166]	MDS-Unet	-	-	-	-	-	81.7	90.3	0.77	-	1.52G	4.0
[167]	Resnet50/MobileNet V2	88.3	-	-	-	98.3	-	-	2.23	-	0.32G	3.5
[168]	ResNet18	74.6	62.8	-	65.9	-	50.8	-	11.7	44.0	-	2.5
[169]	MobileNetV2	88.6	-	-	-	-	-	-	-	3.9	-	2.5
[170]	BF2SkNet + Cubic SVM	97.1	94.9	-	95.0	99.4	-	-	5.3	-	-	2.5
[171]	GSCEU-Net	95.7	87.4	-	-	-	81.5	89.5	0.041	-	0.081G	4.0
[172]	G-DMN	87.1	75.3	96.2	77.3	95.2	-	-	5.33	-	-	3.5
[173]	SqueezeNet-Light	95.6	94.0	96.0	95.2	-	-	-	20.18	9.3	68.3M	4.0
[174]	ResNet-50 based on the involution and Triplet++ Attention	99.4	98.7	-	-	-	-	-	30.5	-	3.07G	3.0
[175]	REDAUNet	94.4	90.9	97.7	-	-	83.2	90.2	47.77	-	-	3.5
[176]	EfficientnetB0+EDS RGAN	99.0	99.0	99.8	99.0	-	-	-	1.52	23.45	-	3.0
[177]	MAAU	95.5	89.3	-	90.0	-	80.9	88.1	4.2	-	-	3.5
[178]	Light-Dermo	99.1	97.4	98.2	98.1	-	-	-	1.9	9.3	67.3M	4.0
[179]	SkinDistilViT	88.5	88.5	-	88.4	-	-	-	43.3	-	-	3.5
[180]	CNN	88.7	-	-	-	-	-	-	11.96	-	-	3.0
[181]	TCC Neural Network	75.0	-	-	55.0	-	-	-	5.6	-	-	2.5
[182]	AViT	-	-	-	-	-	85.2	91.7	99.4	-	20.9G	4.0
[183]	LCNet	90.4	90.4	-	90.4	-	-	-	3.3	-	-	2.5
[184]	Squeeze-MNet	99.4	99.0	-	-	98.9	-	-	6.44	50.44	-	3.5
[185]	HI-MViT	93.2	93.2	-	93.1	97.7	-	-	4.9	-	-	3.0
[186]	SSD MobileNetV2	96.0	-	-	-	-	-	-	2.2	18.0	-	2.5
[187]	DenseNet169	92.3	93.6	-	93.3	-	-	-	14.3	57.0	-	4.0
[188]	EGFNet	93.9	85.0	98.0	-	-	74.5	84.9	0.52	86.9	0.48G	3.5
[189]	EW-FCM + wide- ShuffleNet	84.8	70.7	96.0	72.6	-	-	-	1.8	-	-	3.5
[190]	LCNet	90.4	90.4	90.4	90.4	-	-	-	3.32	-	-	3.5

[191]	AL-Net	93.1	-	-	-	-	79.5	-	3.45	-	-	2.5
[192]	SLS-Net	97.6	87.8	99.9	-	-	82.0	90.6	2.35	-	-	3.5
[193]	MnasNet	83.5	-	-	-	-	-	-	-	12.5	-	3.0

Note: A dash (-) indicates that the specific metric was not evaluated or reported by the original authors. These studies remain fully eligible in the corpus because these papers are successfully reported at least one other primary computational metric required by the inclusion criteria.

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